

C2062 – Anorganická chemie II

Demonstrační pokusy – d-blok – 3.–7. skupina

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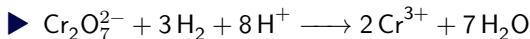
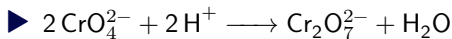
IUPAC Periodic Table of the Elements

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Rb rubidium 85.47																		Sr strontium 87.62																		Y yttrium 88.91																		Zr zirconium 91.22																		Nb niobium 92.91																		Mo molybdenum 95.94																		Tc technetium 98.01																		Ru ruthenium 101.07																		Rh rhodium 102.91																		Pd palladium 106.91																		Ag silver 107.87																		Cd cadmium 112.41																		In indium 114.82																		Sn tin 118.71																		Sb antimony 121.76																		Te tellurium 127.60																		I iodine 126.90																		Xe xenon 131.29																		Cs cesium 132.91																		Ba barium 137.33																		La lanthanum 138.91																		Ce cerium 140.12																		Pr praseodymium 140.91																		Nd neodymium 144.24																		Pm promethium 144.91																		Sm samarium 150.36																		Eu europium 151.96																		Gd gadolinium 157.25																		Tb terbium 158.93																		Dy dysprosium 162.50																		Ho holmium 164.93																		Er erbium 167.26																		Tm thulium 168.93																		Yb ytterbium 173.05																		Lu lutetium 174.97																		Hf hafnium 178.49																		Ta tantalum 180.95																		W tungsten 183.85																		Re rhenium 186.21																		Os osmium 190.23																		Ir iridium 192.22																		Pt platinum 195.08																		Au gold 196.97																		Hg mercury 200.59																		Tl thallium 204.38																		Pb lead 207.2																		Bi bismuth 208.98																		Po polonium 209																		At astatine 210																		Rn radon 222																		Fr francium 223																		Ra radium 226																		Ac actinium 227																		Th thorium 232.04																		Pa protactinium 231.04																		U uranium 238.03																		Np neptunium 237.05																		Pu plutonium 244.06																		Am americium 243.06																		Cm curium 247.07																		Bk berkelium 247.07																		Cf californium 251.08																		Es einsteinium 252.08																		Fm fermium 257.10																		Md mendelevium 258.10																		Lv livermorium 260.10																		Ts tennessine 289.10																		Og oganeson 289.10																	

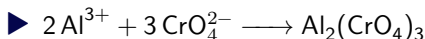
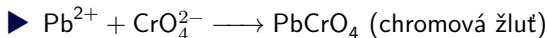
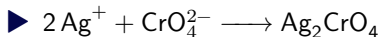


57 La lanthanum 138.90547(3)	58 Ce cerium 140.12(1)	59 Pr praseodymium 140.90768(2)	60 Nd neodymium 144.24(2)	61 Pm promethium 144.9127(2)	62 Sm samarium 150.36(2)	63 Eu europium 151.964(2)	64 Gd gadolinium 157.25(3)	65 Tb terbium 158.925(2)	66 Dy dysprosium 162.50(2)	67 Ho holmium 164.93032(2)	68 Er erbium 167.259(2)	69 Tm thulium 168.93032(2)	70 Yb ytterbium 173.054(2)	71 Lu lutetium 174.967(1)
89 Ac actinium 227.033(2)	90 Th thorium 232.0377(2)	91 Pa protactinium 231.03688(2)	92 U uranium 238.02891(3)	93 Np neptunium 237.04817(3)	94 Pu plutonium 244.0642(2)	95 Am americium 243.06136(2)	96 Cm curium 247.0763(2)	97 Bk berkelium 247.07125(2)	98 Cf californium 251.0832(2)	99 Es einsteinium 252.083(2)	100 Fm fermium 257.1053(2)	101 Md mendelevium 258.1058(2)	102 No nobelium 259.10758(2)	103 Lr lawrencium 262.10536(2)

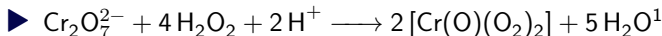
Redukce chromanu vodíkem



Srážení chromanů



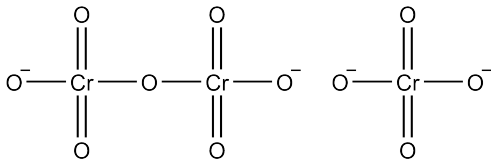
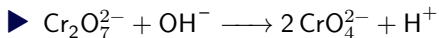
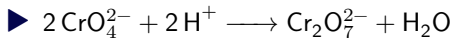
Komplex oxido-diperoxidochromový



¹Peroxokomplex chromu

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Reverzibilní přechod chroman–dichroman



Struktura dichromanu a chromanu

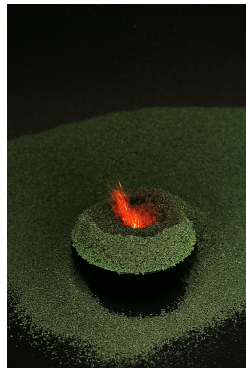
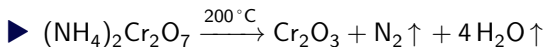


Roztoky dichromanu a chromanu.²

²Zdroj: Capaccio/Commons

Demonstrační pokusy – d-blok – 3.–7. skupina

- ▶ Dichroman amonný podléhá samovolnému rozkladu za uvolnění plynného dusíku a vzniku oxidu chromitého:^{3,4,5}



³Zdroj: Rando Tuvikene/Commons

⁴Zdroj: Rando Tuvikene/Commons

⁵Zdroj: Rando Tuvikene/Commons

Alkalické tavení MnO_2

- ▶ Oxid mangančitý (MnO_2) je možné oxidovat kyslíkem v bazickém prostředí na manganan.
- ▶ $2 \text{KNO}_3 \longrightarrow 2 \text{KNO}_2 + \text{O}_2$
- ▶ $2 \text{MnO}_2 + 4 \text{KOH} + \text{O}_2 \longrightarrow 2 \text{K}_2\text{MnO}_4 + 2 \text{H}_2\text{O}$
- ▶ Ten následně disproportionuje za vzniku manganistanu a oxidu mangančitého.
- ▶ $3 \text{K}_2\text{MnO}_4 + 4 \text{HCl} \longrightarrow 2 \text{KMnO}_4 + \text{MnO}_2 + 4 \text{KCl} + 2 \text{H}_2\text{O}$